

Set Theory

Set theory is a branch of mathematics that deals with collections of objects, called sets. A set is simply a collection of distinct elements, such as numbers, letters, or even everyday objects, that share a common property or rule.

Colors of a Traffic Light



{Red, Yellow, Green}

Letters in the English Alphabet



{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z}

Planets in the Solar System



{Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune}

Example of Collections which are Not Set

- **Beautiful Cities:**

The concept of "beautiful" is subjective and depends on personal opinions.

- **Interesting Books:**

Everyone has a different idea of what is "interesting," so it is not a fixed collection.

- **Best Movies of All Time:**

The idea of "best" varies based on individual preferences, making this collection undefined.

Set Properties

Membership and Equality

Membership is the basic property of sets from which all other operations and properties can be defined. A set either contains an object or does not contain an object. Sets define a clear and unambiguous description of its members.

We have already see the notation $\in$ which is used say that an element is included in the set. The notation $\notin$ says an element is not included in a set.

$$a \in \{a, b, c, d, e\}$$
$$4 \notin \{1, 3, 5, 7, 9, \dots\}$$

Definition (equality of sets)

Two sets A and B are *equal* if every member of A is a member of B and every element of B is a member of A .

$$\{1, 2\} = \{2, 1\} = \{1, 1, 2\} = \{2, 1, 2, 1, 1, 1, 1, 2, 1, 2, 1, 2, 2\}$$

Special Sets: Empty and Universal

Some important sets have special names. In particular, the empty set and the universal set. Thinking again about set membership, their names suggest that these sets contain *nothing* or *everything*.

Definition (empty set)

The *empty set* is the unique set containing no elements. It is denoted by $\emptyset$.

Such a universe can be implicit, for example the set containing days of the week $D = \{\text{Monday, Wednesday, Friday}\}$, implicitly has $U = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$.

Sets and empty sets

Are the following equalities true or false?

1. $\emptyset = \{\}$
2. $\emptyset = \{x \in \mathbb{Z} \mid 2x = 3\}$
3. $\emptyset = \{\{\}\}$
4. $\emptyset = \{x \in \mathbb{R} \mid x^2 + 2 = 0\}$
5. $\emptyset = \{\emptyset\}$

Sets and empty sets

1. **True.**
2. **True.** No integer satisfies the equation $2x = 3$.
3. **False.** The left-hand side is the empty set. The right-hand side is the set *containing* the empty set.
4. **True.** No real number satisfies the equation $x^2 + 2 = 0$.
5. **False.** The left-hand side is the empty set. The right-hand side is the set *containing* the empty set.

Size of a Set

The last basic property of a set is its *size*. Sets are crucial in counting and combinatorics, so being able to discuss their size concretely is important. The terms “cardinality” and “size” are often used interchangeably.

Definition (cardinality)

The *cardinality* of a finite set is the number of distinct elements contained in the set. For a set A its cardinality is denoted $|A|$.

Definition (subset)

A set A is a *subset* of the set B , written $A \subseteq B$, if every member of A is also a member of B .

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Subsets

The following are valid subsets.

$$\{1, 2, 3\} \subseteq \{1, 2, 3, 4, 5\}$$

$$\{c, e\} \subseteq \{a, b, c, e, y, z\}$$

$$\{2, 4, 6\} \subseteq \{2, 4, 6\}$$

$$\{\text{foo}, \text{bar}\} \subseteq \{\text{bar}, \text{buzz}, \text{foo}\}$$

Tip

By the definition of subset, we have $\emptyset \subseteq A$ for any set A . Indeed, since $\emptyset$ has no elements, it is the subset of anything.

Equality as subsets

Let $A = \{1, 3, 5, 7, 9\}$. Let $B = \{3, 5\}$. Let C be the positive odd integers less than 10. Let D be the positive odd integers less than 100.

We have: $B \subseteq A$, $B \subseteq A$, $B \subseteq C$, $B \subseteq D$.

We also have $A \subseteq C$, $A \subseteq D$, $C \subseteq A$, $C \subseteq D$.

Therefore, $A = C$.

- | | |
|-------------------|-------------------------------------|
| • Proper subset | • $A \subset B$ or $A \subsetneq B$ |
| • Superset | • $A \supseteq B$ |
| • Proper superset | • $A \supset B$ or $A \supsetneq B$ |

Definition (power set)

The *power set* of a set A is the set of all possible subsets of A . The power set of A is denoted by $\mathcal{P}(A)$.

A power set

Let $A = \{a, b, c\}$. The members of $\mathcal{P}(A)$ are:

1. $\emptyset$
2. $\{a\}$
3. $\{b\}$
4. $\{c\}$
5. $\{a, b\}$
6. $\{a, c\}$
7. $\{b, c\}$
8. $\{a, b, c\}$

Moreover, $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

Theorem

For a finite set A with $|A| = n$, $|\mathcal{P}(A)| = 2^n$.

Operations on Sets

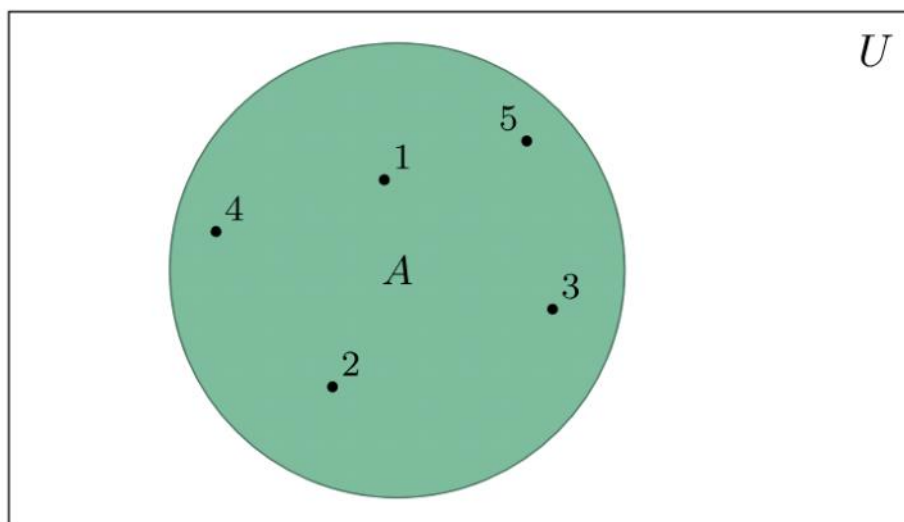


Fig. 2.1 The set $A = \{1, 2, 3, 4, 5\}$ in a universe U represented as a Venn diagram.

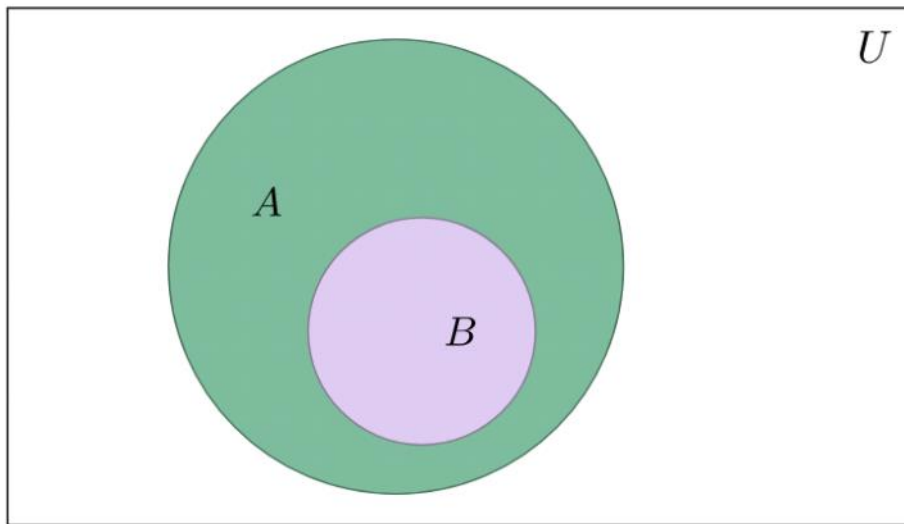


Fig. 2.2 The sets A and B with $B \subseteq A$ in a universe U represented as a Venn diagram.

Definition (set union)

The *union* of two sets A and B is the set of elements which are members of A or members of B , or members of both. The union of A and B is denoted $A \cup B$.

In set builder notation, the union of two sets A and B can be written as:

$$\{x \mid x \in A \vee x \in B\}$$

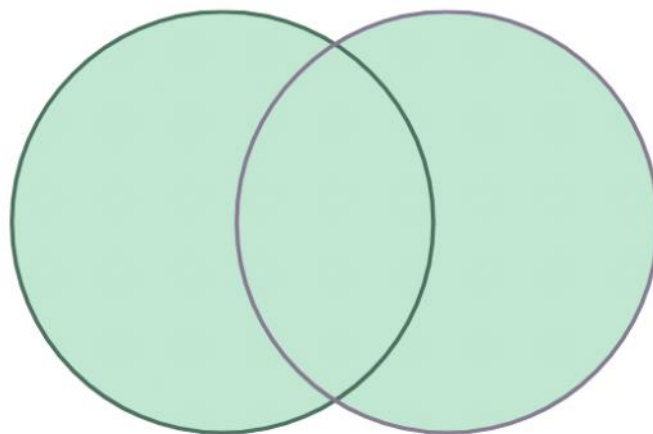


Fig. 2.3 The union of sets A and B is everything in A or B or both.

Definition (set intersection)

The *intersection* of two sets A and B is the set of elements which are members of both A and B . The intersection of A and B is denoted $A \cap B$

In set builder notation, the intersection of two sets A and B can be written as:

$$\{x \mid x \in A \wedge x \in B\}$$

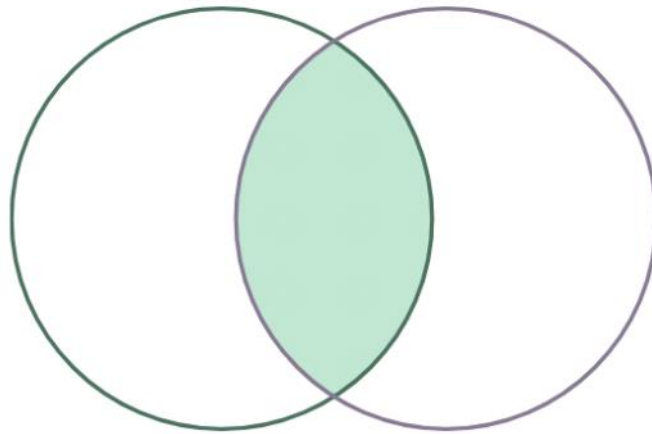


Fig. 2.4 The intersection of sets A and B is members of both A and B .

Union and Intersection

Let $A = \{1, 2, a, b, c\}$ and $B = \{2, 4, 6, x, a\}$.

$$A \cup B = \{1, 2, 4, 5, a, b, c, x\}$$

$$A \cap C = \{2, a\}$$

$$A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

$$A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

Definition (disjoint sets)

Sets A and B are said to be *disjoint* if $A \cap B = \emptyset$.

Set union and intersect

Let:

- $A = \{a, b, y, a, c\}$
- $B = \{4, 5, 6, c, z\}$
- $C = \{a, 2, 3, 4, 5, b\}$
- $D = \{1, 2, 3, 4\}$

What is the result of the following unions and intersections?

1. $A \cup B$
2. $A \cup B \cup C$
3. $A \cup C$
4. $A \cap D$
5. $(A \cup C \cup B) \cap D$

Set union and intersection also have some important identities.

- $A \cup \emptyset = A$
- $A \cap \emptyset = \emptyset$
- $A \cup A = A \cap A = A$
- $(A \cup B) \cap A = A$
- $(A \cap B) \cup A = A$

Proposition

For two sets A and B the cardinality of $A \cup B$ is given by:

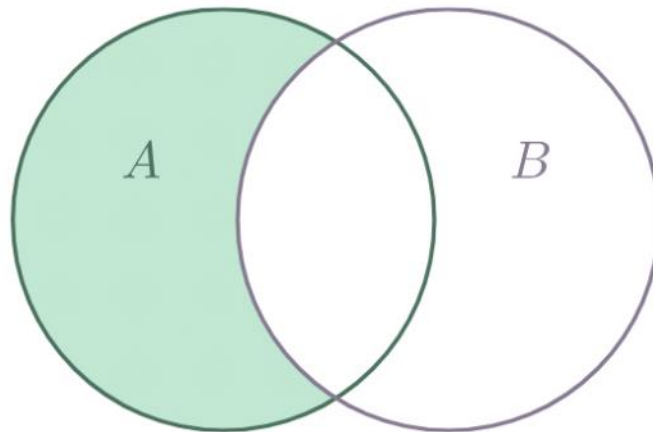
$$|A \cup B| = |A| + |B| - |A \cap B|$$

Definition (set difference)

The *difference* between two sets A and B is the set of elements which are members of A which are not members of B . The difference of A by B is denoted $A \setminus B$.

In set builder notation, the set difference $A \setminus B$ can be written as:

$$\{x \mid x \in A \wedge x \notin B\}$$

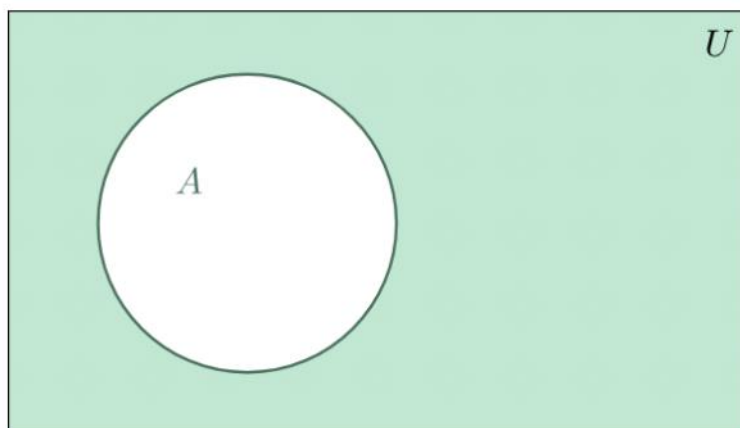


Definition (set complement)

The *complement* of a set A is the set of elements which are not members of A . The set complement is denoted by $\overline{A}$ or A^c .

In set builder notation, the set complement $\overline{A}$ can be written as:

$$\{x \in U \mid x \notin A\}$$



Set difference and complement also have some important identities.

- $A \setminus B = A \cap \overline{B}$
- $(A^c)^c = A$
- $\overline{\emptyset} = U$
- $A \setminus \emptyset = A$
- $A \setminus U = \emptyset$

 Definition (symmetric difference)

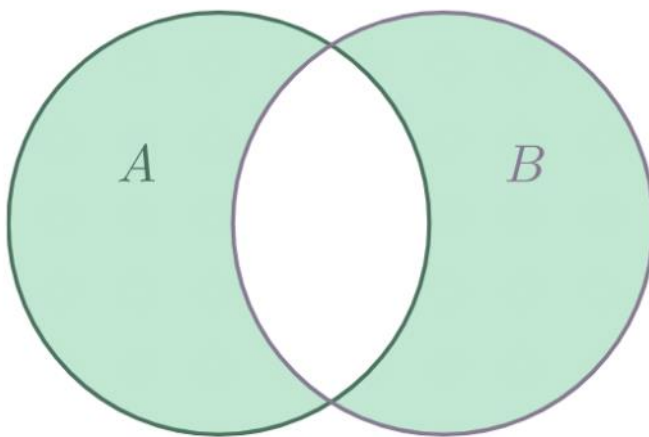
The *symmetric difference* of two sets A and B is the set of elements which are members of A or members of B but not both. The symmetric difference of A and B is denoted $A \oplus B$.

The “symmetric” in symmetric difference comes from the following identity:

$$A \oplus B = (A \setminus B) \cup (B \setminus A)$$

However, a more intuitive way to think about symmetric difference is as the union of two sets minus their intersection.

$$A \oplus B = (A \cup B) \setminus (A \cap B)$$



Cartesian Product

Definition (tuple)

A *tuple* is a finite ordered sequence of elements (from a set). An n -tuple is a tuple with n elements.

- a 1-tuple is a *monuple* or a *singleton*,
- a 2-tuple is a *couple* or a *pair*,
- a 3-tuple is a *triple* or *triplet*,
- a 4-tuple is a *quadruple*,
- a 5-tuple is a *pentuple*, and so on . . .

Tuple equality

The tuples $(3, 7, 5)$ and $(3, 7, 5)$ are equal. However, $(3, 5, 7) \neq (3, 7, 5)$.

Definition (Cartesian product)

The Cartesian product of two sets A and B is the set of all pairs (a, b) where $a \in A$ and $b \in B$. The Cartesian product is denoted $A \times B$.

Cartesian product

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$.

$$A \times B = \{(1, x), (2, x), (3, x), (1, y), (2, y), (3, y)\}$$

$$B \times A = \{(x, 1), (y, 1), (x, 2), (y, 2), (x, 3), (y, 3)\}$$

If $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$ we have:

	x	y	z
1	$(1, x)$	$(1, y)$	$(1, z)$
2	$(2, x)$	$(2, y)$	$(2, z)$
3	$(3, x)$	$(3, y)$	$(3, z)$

Proving subsets

Prove the following proposition.

Proposition

If $A \subseteq B$ then $A \cup B = B$.

Proving subsets

We look to prove $A \subseteq B \rightarrow A \cup B = B$

Proof:

Assume $A \subseteq B$. We look to prove $x \in A \cup B \leftrightarrow x \in B$

1. $(\rightarrow)$ Assume $x \in A \cup B$. Then $x \in A$ or $x \in B$. Proceed by cases:
 - If $x \in B$, we are done.
 - If $x \in A$, we know $A \subseteq B$, therefore $x \in B$.
2. $(\leftarrow)$ Assume $x \in B$. Therefore, x must surely be in $A \cup B$. ■

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

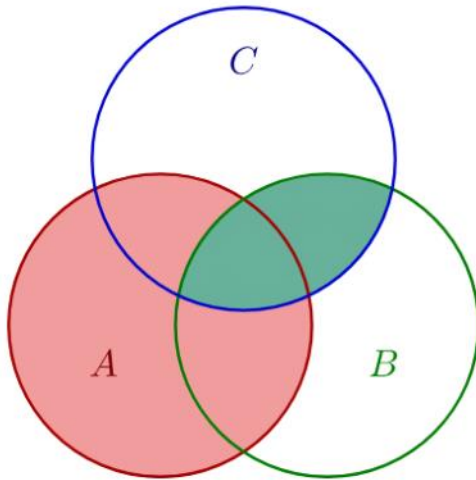


Fig. 2.10 $A \cup (B \cap C)$

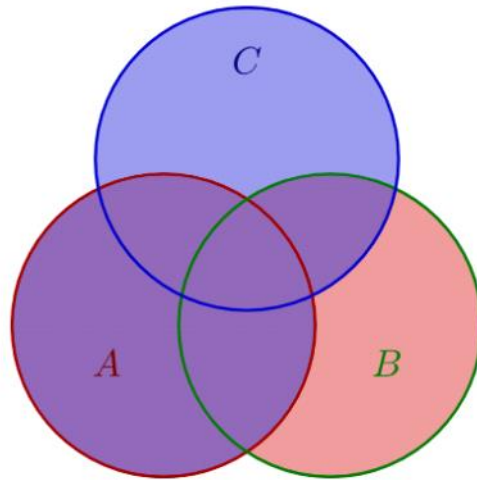


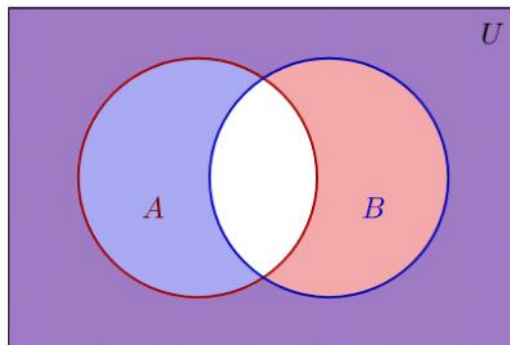
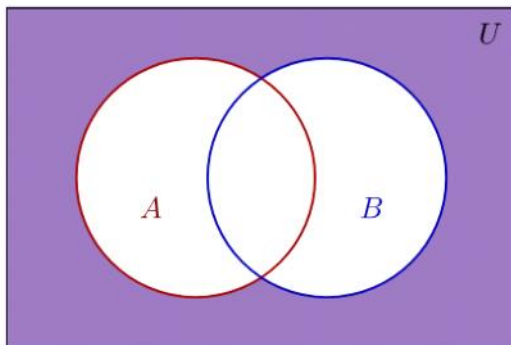
Fig. 2.11 $(A \cup B) \cap (A \cup C)$.
 $A \cup B$ is shown in red and purple,
 $A \cup C$ in blue and purple, and
 $(A \cup B) \cap (A \cup C)$ in purple.

De Morgan's Laws

For sets, we have the following identities.

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$



De Morgan's laws for sets

Draw two Venn diagrams showing $\overline{A \cap B}$ and $\bar{A} \cup \bar{B}$. Convince yourself that these are the same sets.

Exercise 2.5

Let $A = \{2, 3, 5, 7, 9\}$, $B = \{1, 3, 5, 6, 8\}$ and $C = \{x \in \mathbb{Z}_{\geq 0} \mid x < 8\}$. Let $U = \{x \in \mathbb{Z} \mid 0 \leq x \leq 12\}$.

Write out the following sets using the roster method.

1. $(A \cap B) \cup C$
2. $A \setminus C$
3. $(B \cup C) \setminus A$
4. $(B \cap C) \setminus C$
5. $\overline{A} \cup B$

Prove the following proposition.

Proposition

If $A \subseteq B$ then $A \cap B = A$.

Prove the following proposition.

Proposition

Let A , B , and C be sets. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Prove the following proposition.

Proposition

For any two sets A and B , $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Relations and Functions

- A **relation** is simply a connection between two sets of objects. Think of it as a rule that pairs elements from one set with elements from another set.

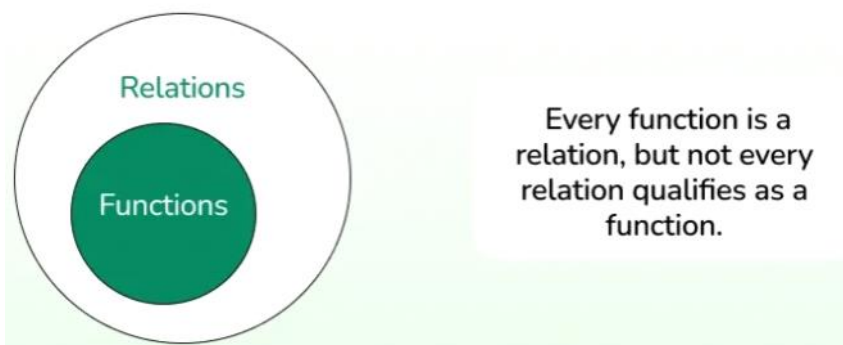
For example, the relationship "is greater than" pairs numbers like 5 and 3 because 5 is greater than 3.

- A **function**, on the other hand, is a special kind of relation. In a function, each element from the first set (called the domain) is connected to exactly one unique element in the second set (called the range).

For instance, if you assign a name to each roll number in a class, that is a function because each roll number is paired with one unique name.

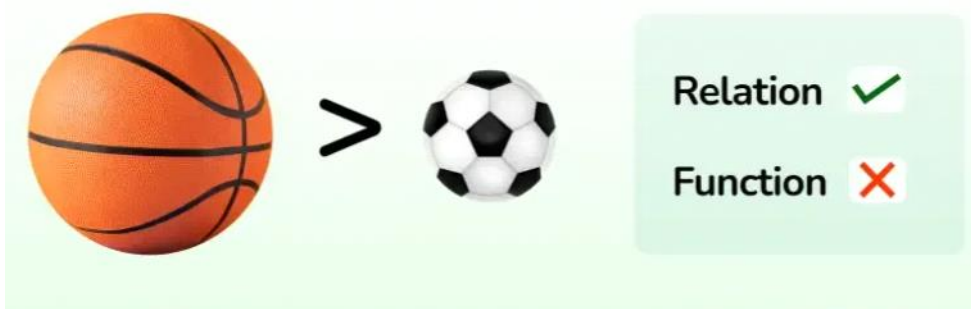
DEFINITION A **function** f from a set D to a set Y is a rule that assigns a *unique* (single) element $f(x) \in Y$ to each element $x \in D$.

Relations And Functions



Relation and Function

Example: is greater than



Relation and Function

Example: is twice of

$$8 = 2 \times 4$$

Relation ✓

Function ✓

Relation and Function

Example: $f: (1,8), (2,6), (4,4), (2,7), (5,9)$

$$f(2) = 6$$

$$f(2) = 7$$

Relation ✓

Function ✗

Relation and Function

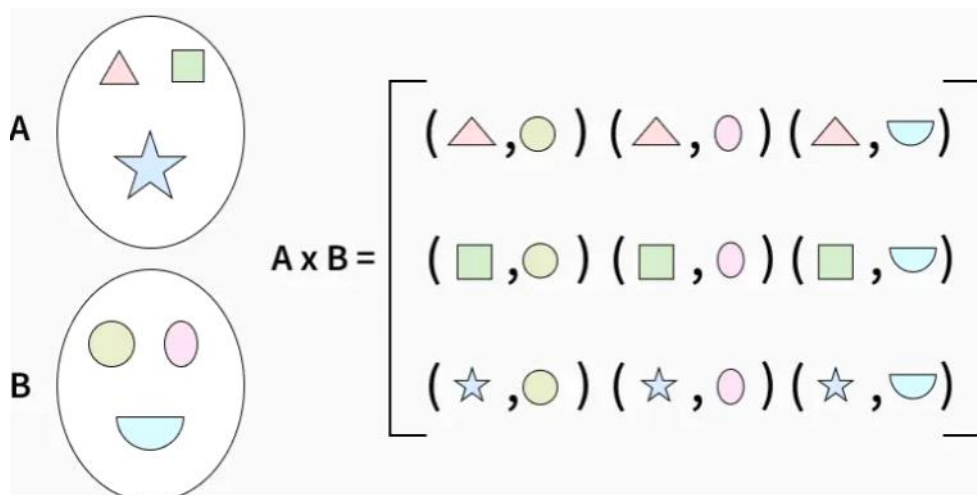
Example: $f(x) = |x|$

$$f(-1) = 1$$

$$f(1) = 1$$

Relation ✓

Function ✓



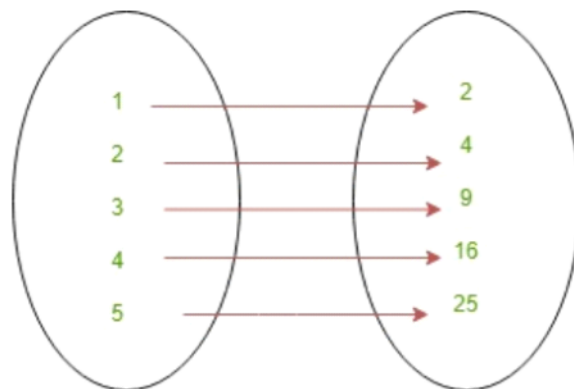
B	x	y	z	A
	(1,x)	(1,y)	(1,z)	B x A
	(2,x)	(2,y)	(2,z)	
	(3,x)	(3,y)	(3,z)	

Solved Problems on Cartesian Product of Sets

Problem 1. If $A = \{9, 10\}$ and $B = \{3, 4, 6\}$, find $A \times B$ and $|A \times B|$?

Example: $\{(1, 1), (2, 4), (3, 9), (4, 16), (5, 25)\}$

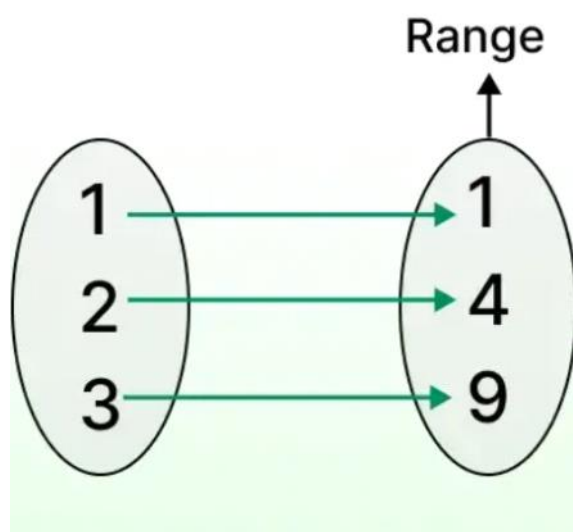
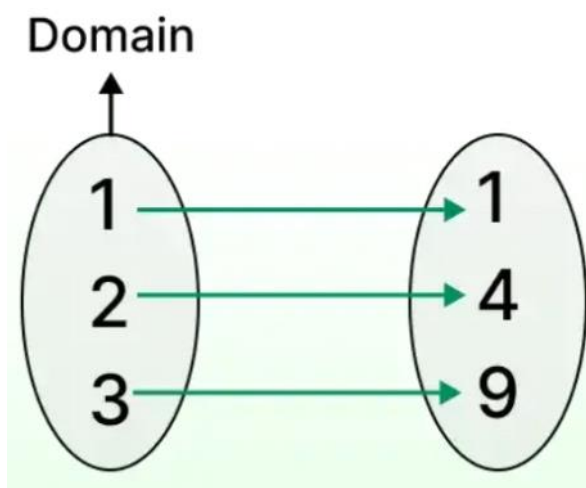
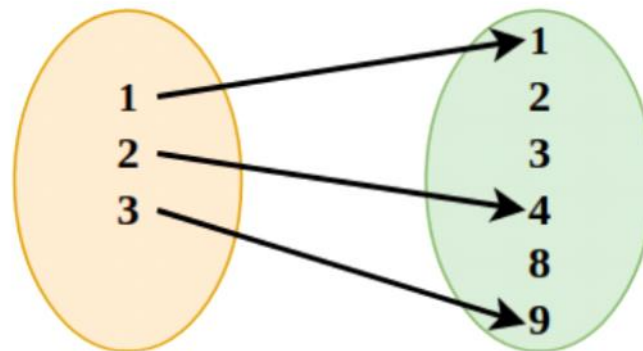
This represent square of a number which means if $x=1$ then $y = x \cdot x = 1$ and so on.

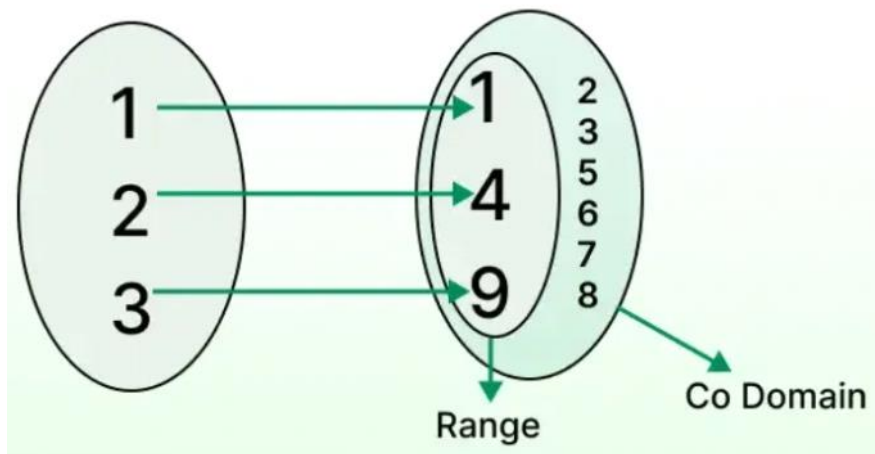


Domain and Range of Relations

The **domain** is the set of all possible input values (the "x" values), and the **range** is the set of all possible output values (the "y" values) in a relation.

Example: The relation $R = \{(1, 1), (2, 4), (3, 9)\}$ is represented using the following diagram:





Valid pairs we found: (4, 6), (9, 11), (10, 12), (11, 13)

$R = \{(4, 6), (9, 11), (10, 12), (11, 13)\}$

Thus,

- *Domain of R is (4, 9, 10, 11)*
- *Range of R is (6, 11, 12, 13)*
- *Codomain is (4, 5, 6, 9, 10, 11, 12, 13, 17)*

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

DEFINITIONS Let f be a function defined on an interval I and let x_1 and x_2 be any two points in I .

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be **increasing** on I .
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be **decreasing** on I .

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DEFINITIONS A function $y = f(x)$ is an

even function of x if $f(-x) = f(x)$,

odd function of x if $f(-x) = -f(x)$,

for every x in the function's domain.

Common Functions

A variety of important types of functions are frequently encountered in calculus. We identify and briefly describe them here.

Linear Functions A function of the form $f(x) = mx + b$, for constants m and b , is called a **linear function**. Figure 1.14a shows an array of lines $f(x) = mx$ where $b = 0$, so these lines pass through the origin. The function $f(x) = x$ where $m = 1$ and $b = 0$ is called the **identity function**. Constant functions result when the slope $m = 0$ (Figure 1.14b). A linear function with positive slope whose graph passes through the origin is called a *proportionality* relationship.

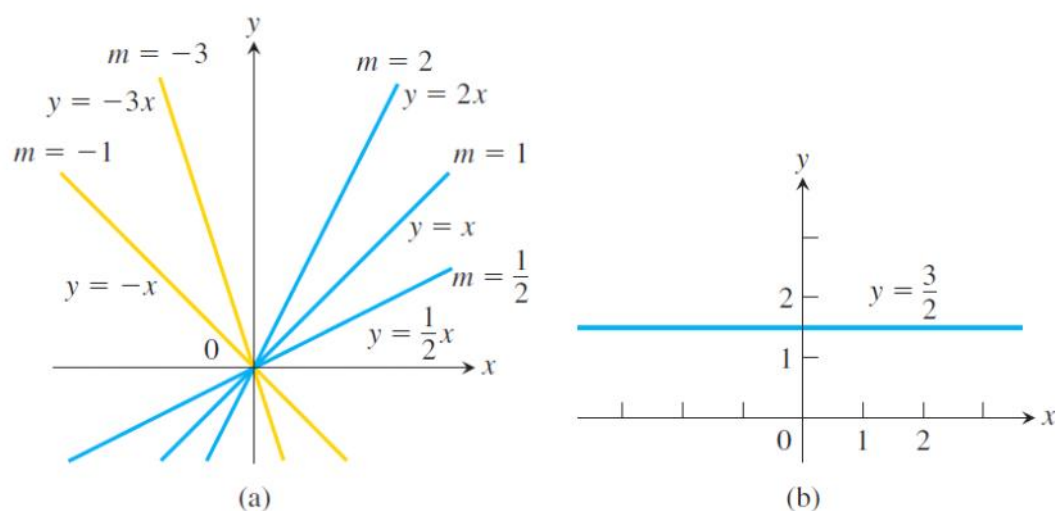


FIGURE 1.14 (a) Lines through the origin with slope m . (b) A constant function with slope $m = 0$.

Power Functions A function $f(x) = x^a$, where a is a constant, is called a **power function**. There are several important cases to consider.

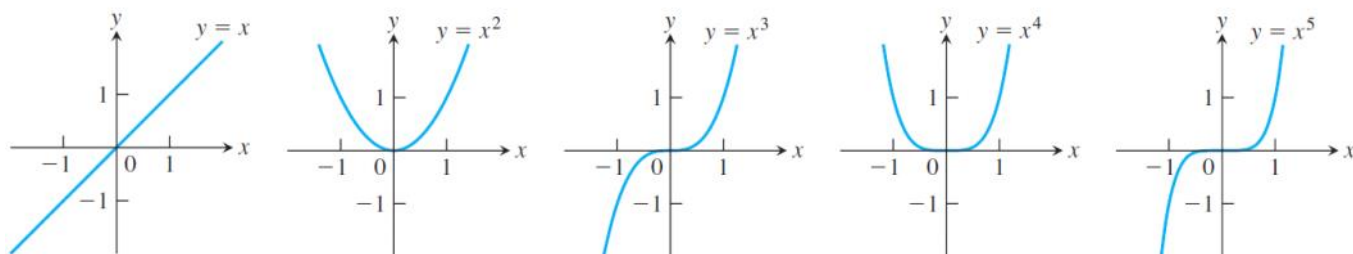


FIGURE 1.15 Graphs of $f(x) = x^n$, $n = 1, 2, 3, 4, 5$, defined for $-\infty < x < \infty$.

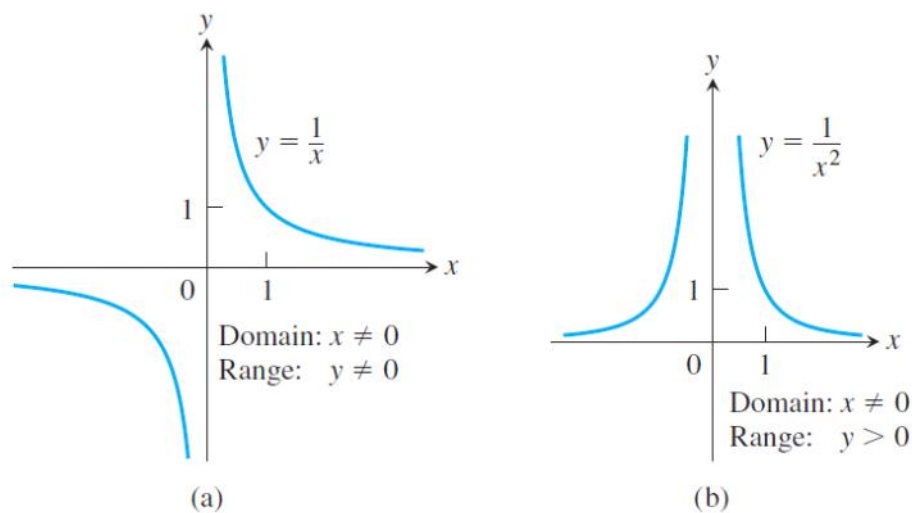
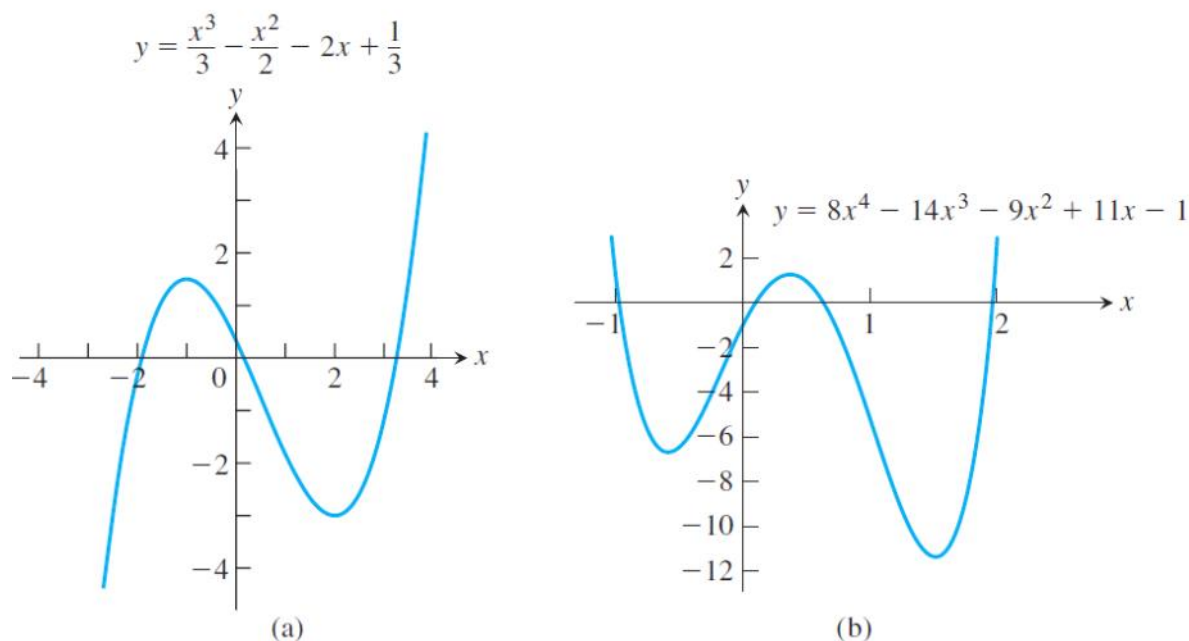


FIGURE 1.16 Graphs of the power functions $f(x) = x^a$ for part (a) $a = -1$ and for part (b) $a = -2$.

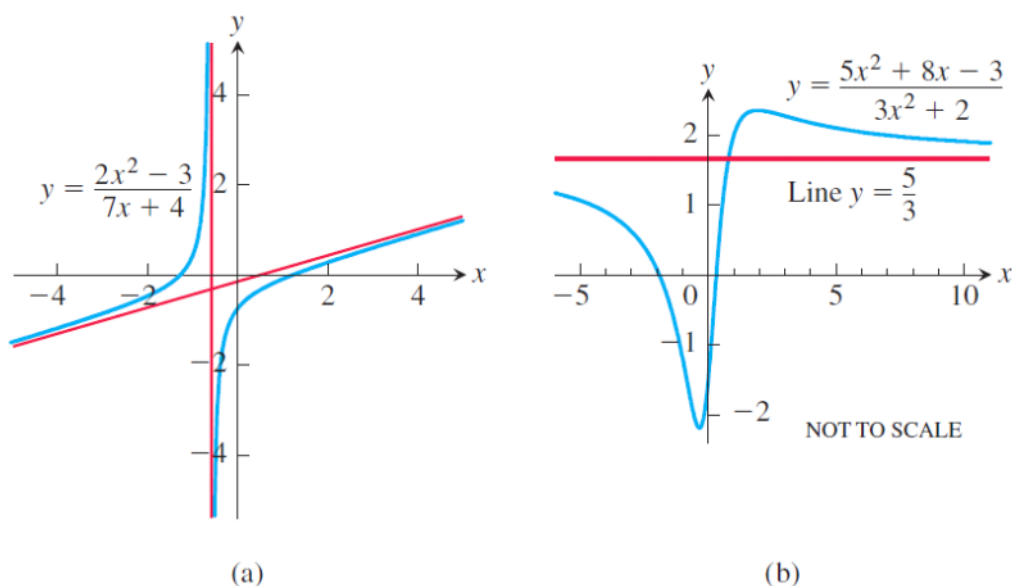
Polynomials A function p is a **polynomial** if

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

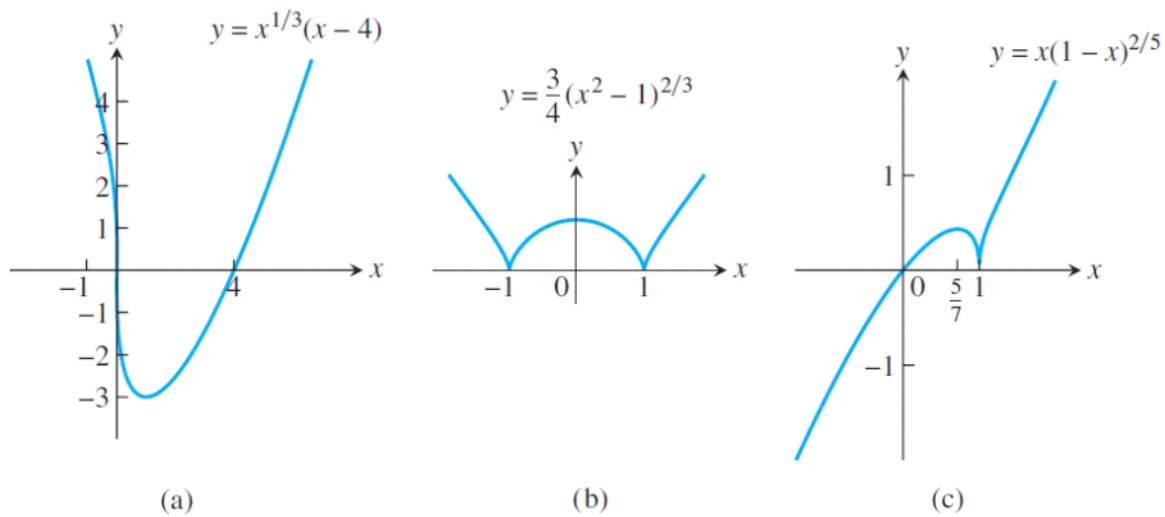
where n is a nonnegative integer and the numbers $a_0, a_1, a_2, \dots, a_n$ are real constants (called the **coefficients** of the polynomial). All polynomials have domain $(-\infty, \infty)$. If the



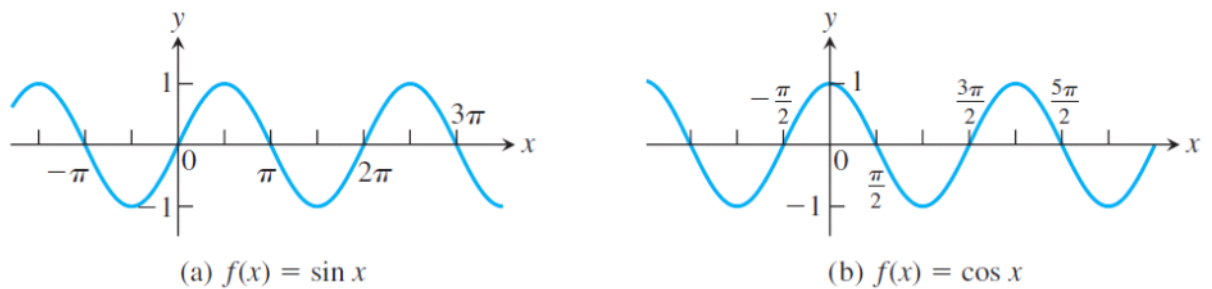
Rational Functions A **rational function** is a quotient or ratio $f(x) = p(x)/q(x)$, where p and q are polynomials. The domain of a rational function is the set of all real x for which $q(x) \neq 0$. The graphs of several rational functions are shown in Figure 1.19.



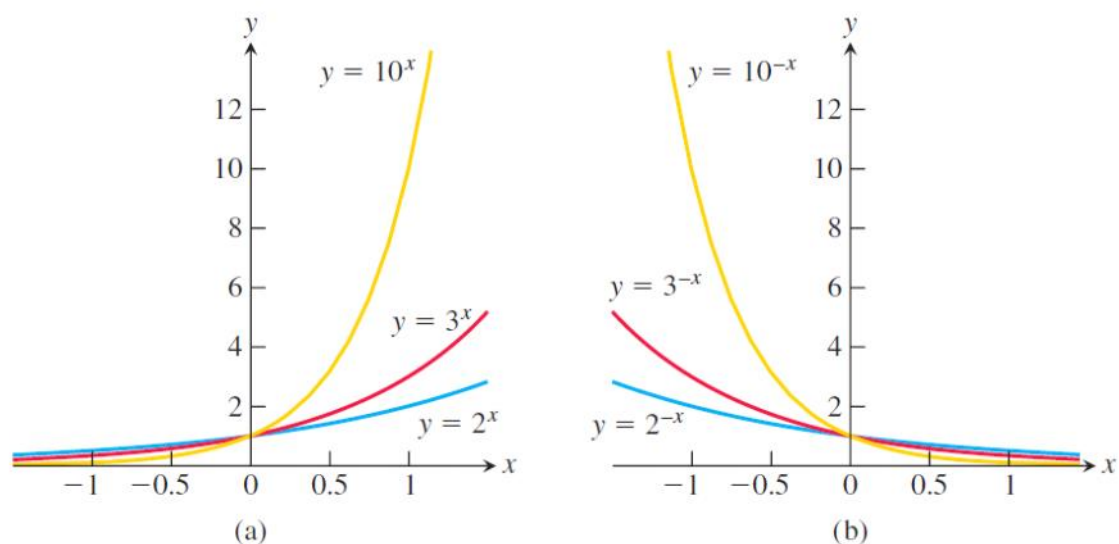
Algebraic Functions Any function constructed from polynomials using algebraic operations (addition, subtraction, multiplication, division, and taking roots) lies within the class of **algebraic functions**. All rational functions are algebraic, but also included are more complicated functions (such as those satisfying an equation like $y^3 - 9xy + x^3 = 0$, studied in Section 3.7). Figure 1.20 displays the graphs of three algebraic functions.



Trigonometric Functions The six basic trigonometric functions are reviewed in Section 1.3. The graphs of the sine and cosine functions are shown in Figure 1.21.



Exponential Functions Functions of the form $f(x) = a^x$, where the base $a > 0$ is a positive constant and $a \neq 1$, are called **exponential functions**. All exponential functions have domain $(-\infty, \infty)$ and range $(0, \infty)$, so an exponential function never assumes the value 0. We discuss exponential functions in Section 1.5. The graphs of some exponential functions are shown in Figure 1.22.



Logarithmic Functions These are the functions $f(x) = \log_a x$, where the base $a \neq 1$ is a positive constant. They are the *inverse functions* of the exponential functions, and we discuss these functions in Section 1.6. Figure 1.23 shows the graphs of four logarithmic functions with various bases. In each case the domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.

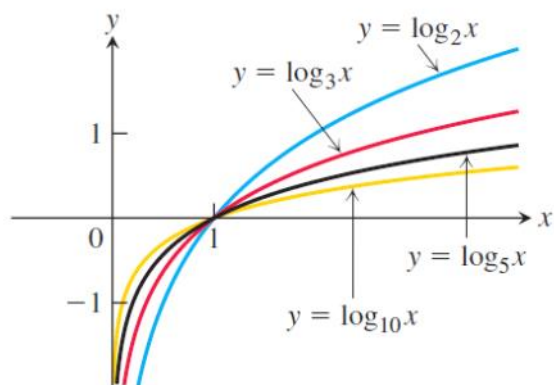


FIGURE 1.23 Graphs of four logarithmic functions.

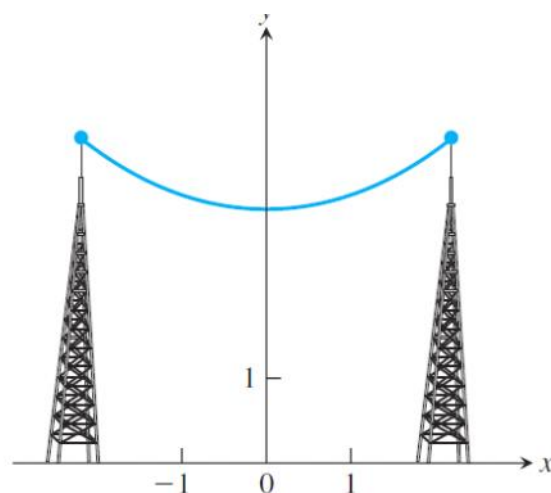


FIGURE 1.24 Graph of a catenary or hanging cable. (The Latin word *catena* means “chain.”)

Transcendental Functions These are functions that are not algebraic. They include the trigonometric, inverse trigonometric, exponential, and logarithmic functions, and many other functions as well. A particular example of a transcendental function is a **catenary**. Its graph has the shape of a cable, like a telephone line or electric cable, strung from one support to another and hanging freely under its own weight (Figure 1.24). The function defining the graph is discussed in Section 7.3.

Functions

In Exercises 1–6, find the domain and range of each function.

1. $f(x) = 1 + x^2$

2. $f(x) = 1 - \sqrt{x}$

3. $F(x) = \sqrt{5x + 10}$

4. $g(x) = \sqrt{x^2 - 3x}$

5. $f(t) = \frac{4}{3 - t}$

6. $G(t) = \frac{2}{t^2 - 16}$

Finding Formulas for Functions

9. Express the area and perimeter of an equilateral triangle as a function of the triangle's side length x .
10. Express the side length of a square as a function of the length d of the square's diagonal. Then express the area as a function of the diagonal length.
11. Express the edge length of a cube as a function of the cube's diagonal length d . Then express the surface area and volume of the cube as a function of the diagonal length.

Piecewise-Defined Functions

Graph the functions in Exercises 25–28.

25. $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2 \end{cases}$

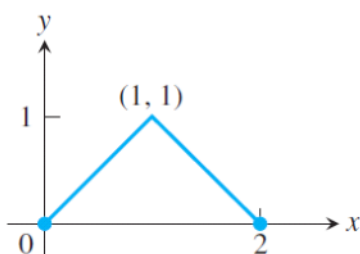
26. $g(x) = \begin{cases} 1 - x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2 \end{cases}$

27. $F(x) = \begin{cases} 4 - x^2, & x \leq 1 \\ x^2 + 2x, & x > 1 \end{cases}$

28. $G(x) = \begin{cases} 1/x, & x < 0 \\ x, & 0 \leq x \end{cases}$

Find a formula for each function graphed in Exercises 29–32.

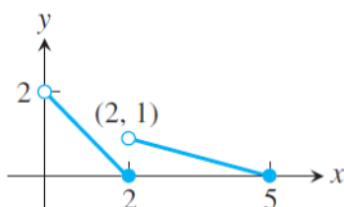
29. a.



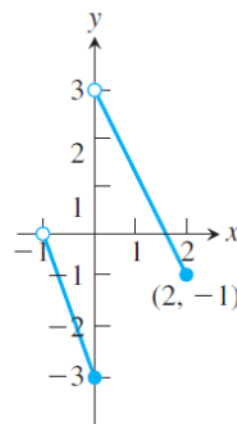
b.



30. a.



b.



- 60. Kinetic energy** The kinetic energy K of a mass is proportional to the square of its velocity v . If $K = 12,960$ joules when $v = 18$ m/sec, what is K when $v = 10$ m/sec?
- 61.** The variables r and s are inversely proportional, and $r = 6$ when $s = 4$. Determine s when $r = 10$.
- 62. Boyle's Law** Boyle's Law says that the volume V of a gas at constant temperature increases whenever the pressure P decreases, so that V and P are inversely proportional. If $P = 14.7$ lb/in² when $V = 1000$ in³, then what is V when $P = 23.4$ lb/in²?
- 63.** A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 14 in. by 22 in. by cutting out equal squares of side x at each corner and then folding up the sides as in the figure. Express the volume V of the box as a function of x .

